

[Get a quote today \(/quote\)](#)

Blog

The Future of the Battery

CHAZ WILKE ([HTTPS://SOLARMEHOME.COM/AUTHOR/CHAZW/](https://solarmehome.com/author/chazw/)) / MAY 4, 2015 ([HTTPS://SOLARMEHOME.COM/THE-FUTURE-OF-THE-BATTERY/](https://solarmehome.com/the-future-of-the-battery/)) / [LEAVE A COMMENT](#) ([HTTPS://SOLARMEHOME.COM/THE-FUTURE-OF-THE-BATTERY/#RESPOND](https://solarmehome.com/the-future-of-the-battery/#respond))



Solar panel technology continues to advance. New developments in solar energy conversion grab headlines daily. Energy conversion rates increase regularly, but there is a limit to what we can do with solar panels until battery technology improves as well. Batteries have remained relatively unchanged for decades, but all that is about to change.

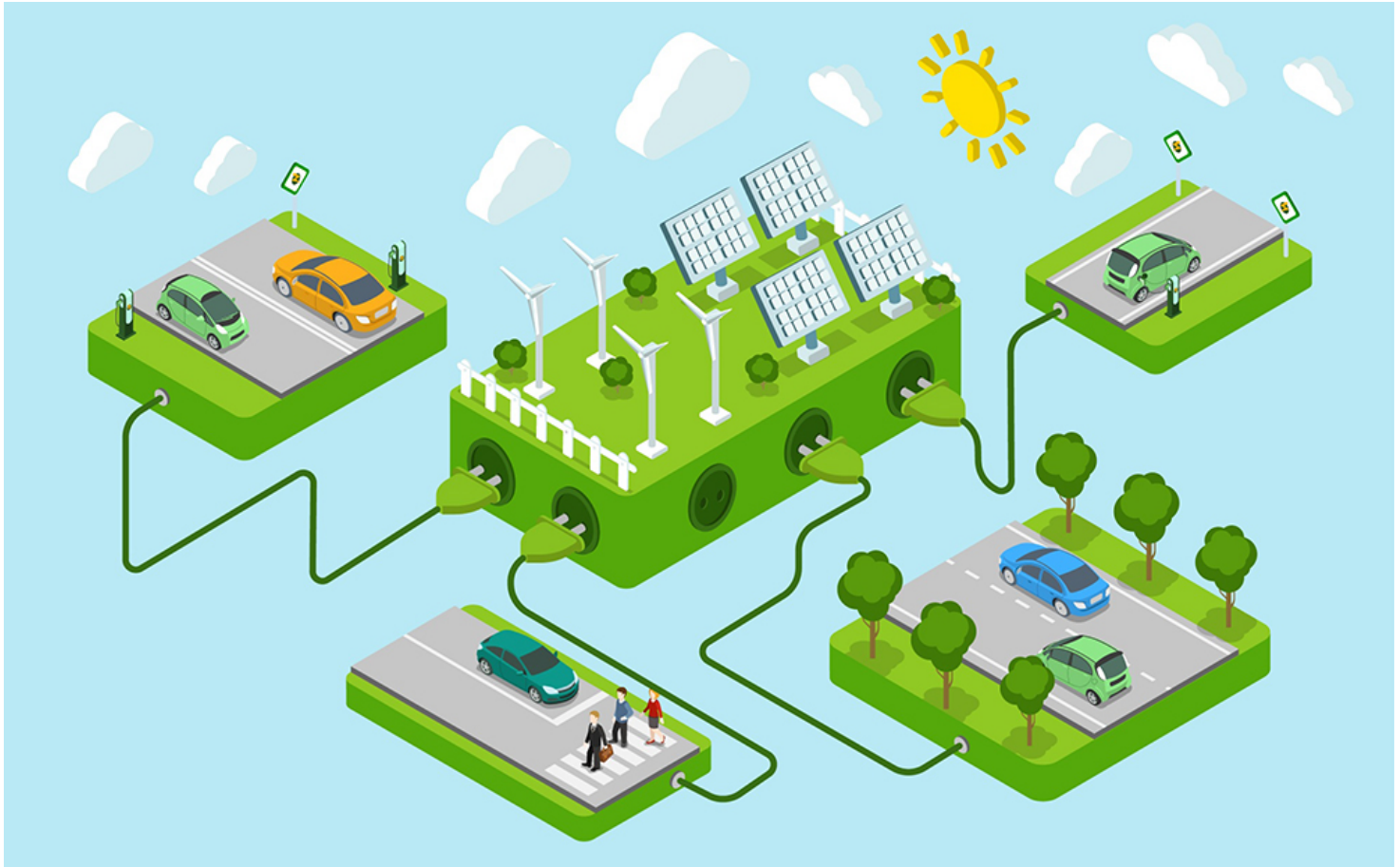
The Battery Conundrum

Consider the size of your phone. It doesn't matter if you have the latest and greatest smartphone, an old flip phone, or something in between. If you look at the size of your phone, the majority of what's inside is battery.

An iPhone has some of the most advanced computer technologies inside the svelte millimeters-thick brick. But, if you were to crack open the phone, you'd see an incredibly small circuit board next to a comparably massive battery. The argument is that you need that amount of power to keep your phone operating all day.

But what if battery capacity wasn't so directly tied to battery size? What if the battery became the smallest component in your phone? And what if there was a battery powerful enough and small enough to retain solar generated power for homes to use when it's cloudy or at nighttime?

We may be getting answers to these questions sooner than you might think.



Emerging Battery Breakthroughs

Wood battery

How on earth could wood become a battery? It can't conduct electricity! You are right to question the logic of wood batteries as we have historically looked to caustic chemicals to store and conduct energy. Alkaline, lithium ion, and nickel cadmium are all caustic. So why are researchers looking at wood as a potential light weight battery solution?

It's simple, really. Trees.

Have you noticed the similarity between photovoltaic solar panel cells and leaves on a tree? It's true we have yet to crack synthetic photosynthesis, but photovoltaic is a good first step. And the second step in that logical progression is identifying how trees store that energy. Surprise, surprise, it's in the tree's wood. "The researchers noted that wood fibers are naturally designed to hold mineral-rich water, water that is very similar (<http://bit.ly/1F21xwP>) to the electrolyte in batteries," explains CleanTechnica.

Wood batteries can be significantly lighter and more pliable than the rigid, heavy contemporary battery counterparts. And the experimental wood batteries are already showing great promise in charge cycles. The prototype is already “successfully lasting through a full 400 charge-discharge cycles.” For comparison, lithium ion batteries, similar to the batteries you have in your phone, last through about 300-500 charge-discharge cycles (<http://bit.ly/1JMbdwq>).

3D printed battery

You’re likely familiar with the concept of 3D printing. This process, similar to a standard printer, prints layers upon layers of a substance to create a three-dimensional object.

What started as an at-home hobbyist activity by printing simple trinkets is now evolving to print more complex items. Like batteries.

Researchers from Lawrence Livermore National Laboratory are using chemical vapor deposition which guides layers of materials into position. Using carbon nanotube aerogels or better known as “hard smoke (<http://bit.ly/1bXcSB5>)” to print a tight graphene geometric design, the researchers are able to make a structure that may one day function as a battery.

How we 3D-print aerogel



The old guard in the battery game, lithium ion, has a tendency to degrade quickly. As evidenced by the 300-500 charge-discharge cycle lifetime limit on most conventional batteries. Graphene may help the stability of these batteries. So while a 3D printed graphene aerogel battery may be years off. Graphene-integrated batteries are already in development at Tesla.



Tesla's game-changing Powerwall home battery

Big news arrived out of California in the waning final hours of April. Elon Musk took to the stage to announce a much-needed missing piece to the solar energy puzzle.

"We have this handy fusion reactor in the sky called the Sun," Elon said to a rowdy crowd during his Keynote (<http://bit.ly/1DPAw8x>) Wednesday night. "We don't have to do anything, it just works. [It] shows up every day, and produces ridiculous amounts of power."

This is information many of us may already know. However we are aware of the current limitations of solar panels. Chief among them is that they don't work so great (read: at all) at night. So maybe we could use rechargeable batteries to harness the excess power generated during the day for use when it's cloudy or at night. But so far there hasn't been much of a ready-made product that would tackle that problem.

"The issue with existing batteries is that they suck... They're expensive, they're unreliable, they're sort of stinky, ugly, bad in every way. They're very expensive... There's not one integrated place you can go and buy a battery that just works, which is what people really want to buy," Musk told the packed auditorium. "That's the missing piece, that's the thing that's needed to have a proper transition to a sustainable energy world."

And Musk believes he's found a solution to that problem. That missing piece.

You may recall news of Tesla teaming up with Panasonic to create the massive Nevada Gigafactory. The zero-energy footprint factory is deep in construction and looking to start operations in 2016 (<http://bit.ly/1bXduqa>). Once operational, Musk predicts the factory's output will be 50 GWh in annual battery production by 2020.

Tesla acknowledges that there's a problem with current Feed-In Tariff plans (<http://bit.ly/1AAczSD>) across the nation. The grid is old, like over a century old in many cities. The wiring cannot handle a rush of additional energy sent back to the utilities. This means that as more consumers adopt rooftop solar panels for personal energy production, the more strain selling back excess energy will cause.

Tesla plans to release home-grade, business-grade and utility-grade battery solutions to contain all that excess energy and prevent grid overload.

The home-grade Tesla Energy Powerwall battery comes with a capacity of 10KWh starting at \$3,500. That sum is the cost to the installer not including the subsidies that may very well decrease the cost to the end user.

Bloomberg Business covers the long strokes of the April 30, Elon Musk announcement below:

Tesla Brings Battery Power Home: What You Need to ...



Takeaway

Battery technology lagged behind computational processor advancements for decades. Now it appears that is set to change. The next two decades will bring battery advancements closer to the future promised by computer processing power. It's not clear which new battery technology will become the industry standard, but it should be comforting to know there are multiple highly advanced technologies already in the research phase and in the case of Tesla's Powerwall, available for order.

DISCLAIMER: Availability of any state, federal, or local programs will vary and are subject to change. Consult with a financial professional before making any decisions. Energy and financial savings are not guaranteed and will vary from customer to customer. The foregoing is provided for informational purposes only and is not expert or professional advice, and should not be relied upon as such. SolarMeHome does not make any recommendations for any particular solar installer or product. SolarMeHome receives payment from participating installers for customer referrals.

◀ BATTERY ▶ TECHNOLOGY ▶ TESLA

